

Technology Stack

Date	04 October 2026
Team ID	POCKETSMART-TEAM
Project Name	PocketSmart AI – Smart Budget & Recommendation Assistant using Gemini Models
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript (static/js/app.js), Jinja2 templates, Font Awesome icons	Lightweight server-rendered pages with small shared JavaScript helpers.
2	Backend / Server-Side	Python, FastAPI, Uvicorn, Pydantic	Async routing, request validation and automatic API documentation.
3	AI Layer	Google Gemini API (google-generativeai), Pillow	Generates structured budget plans and analyses the uploaded outfit image.
4	Data Storage	In-memory Python structures; uploaded images in static/uploads/	Simple storage suited to a local prototype.
5	Security / Authentication	passlib + bcrypt, python-jose (JWT, HS256), httpOnly cookie	Protects accounts and restricts planner and history routes.
6	Testing	pytest, FastAPI TestClient, unittest.mock	Runs the whole API without a Gemini key by mocking AI calls.
7	Development Tools	Git, GitHub, VS Code, Python virtual environment, python-dotenv	Collaboration and reproducible local setup.